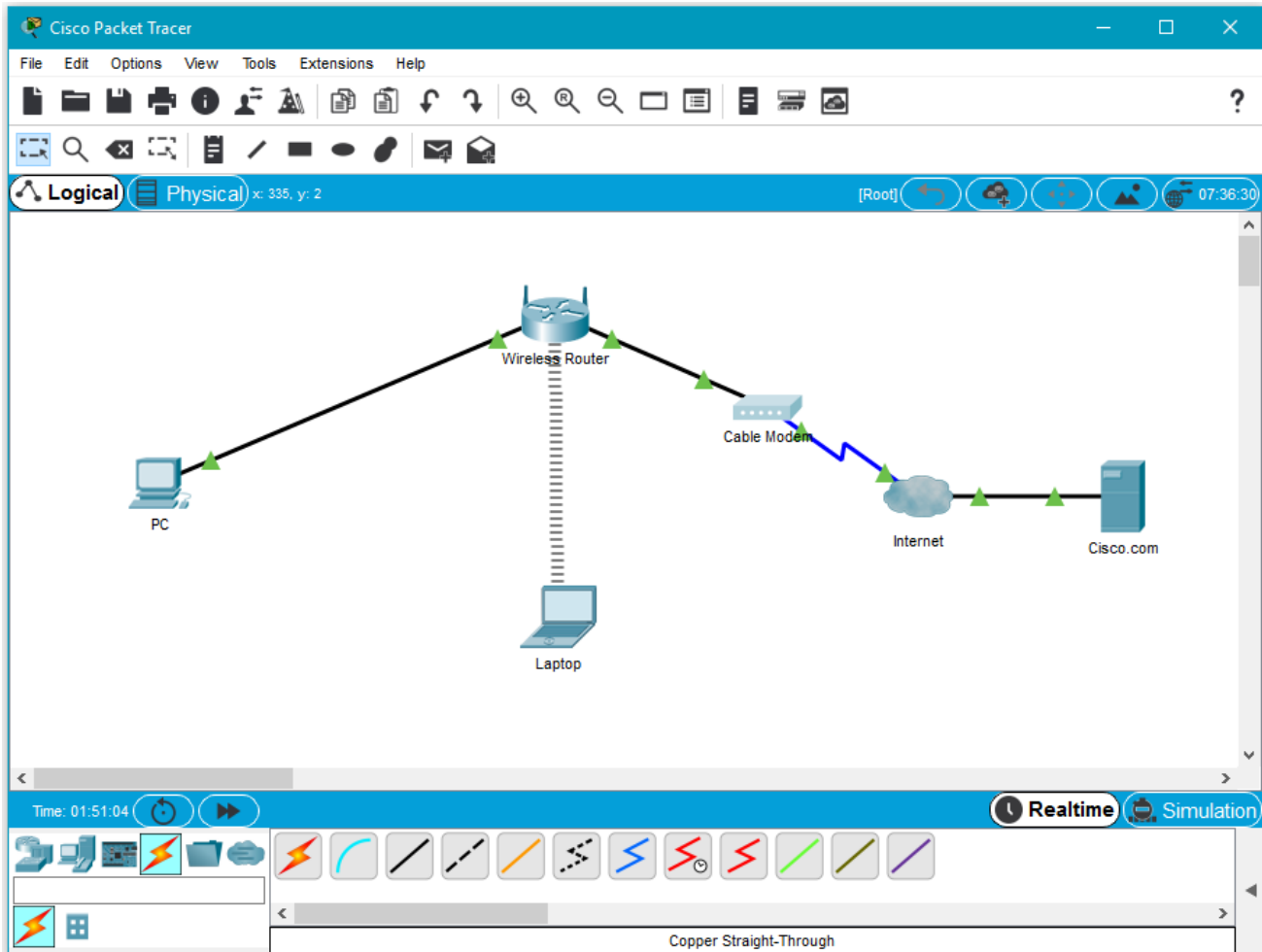


Lab 5: Create a Simple Network Using Packet tracer

Packet Tracer – Create a Simple Network Using Packet Tracer

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC	Ethernet0	DHCP		192.168.0.1
Wireless Router	LAN	192.168.0.1	255.255.255.0	
Wireless Router	Internet	DHCP		
Cisco.com Server	Ethernet0	208.67.220.220	255.255.255.0	
Laptop	Wireless0	DHCP		

Objectives

Part 1: Build a Simple Network in the Logical Topology Workspace

Part 2: Configure the Network Devices

Part 3: Test Connectivity between Network Devices

Part 4: Save the File and Close Packet Tracer

Background / Scenario

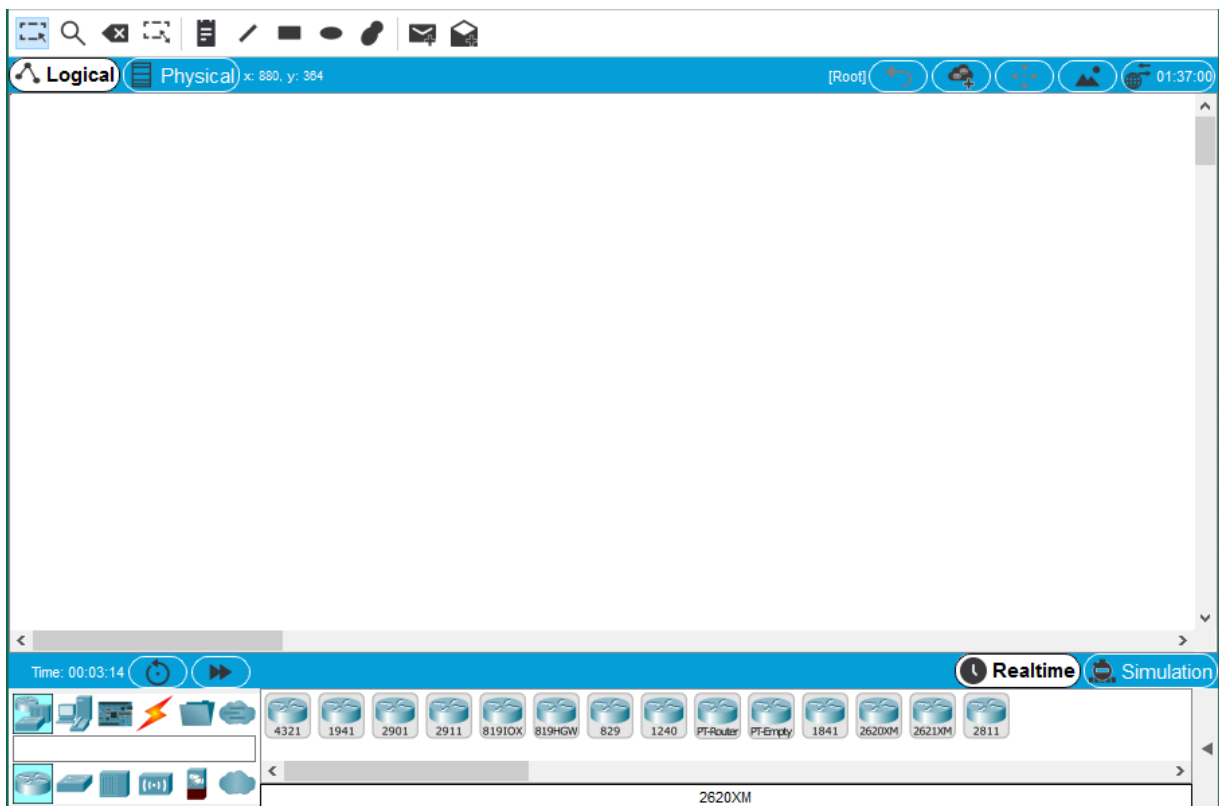
In this activity you will build a simple network in Packet Tracer from scratch and then save the network as a Packet Tracer Activity File (.pkt).

Part 1: Build a Simple Network in the Logical Topology Workspace

Step 1: Launch Packet Tracer.

- a. Launch Packet Tracer on your PC or laptop computer

Double click on the **Packet Tracer** icon on your desktop or navigate to the directory that contains the Packet Tracer executable file and launch Packet Tracer. Packet Tracer should open with a blank default **Logical** topology workspace as shown in the figure.



Step 2: Build the topology

- a. Add network devices to the workspace.

Using the device selection box, add the network devices to the workspace as shown in the topology diagram.

To place a device onto the workspace, first choose a device type from the **Device-Type Selection** box. Then, click on the desired device model from the **Device-Specific Selection** box. Finally, click on a location in the workspace to put your device in that location. If you want to cancel your selection, click the **Cancel** icon for that device. Alternatively, you can click and drag a device from the **Device-Specific Selection** box onto the workspace.

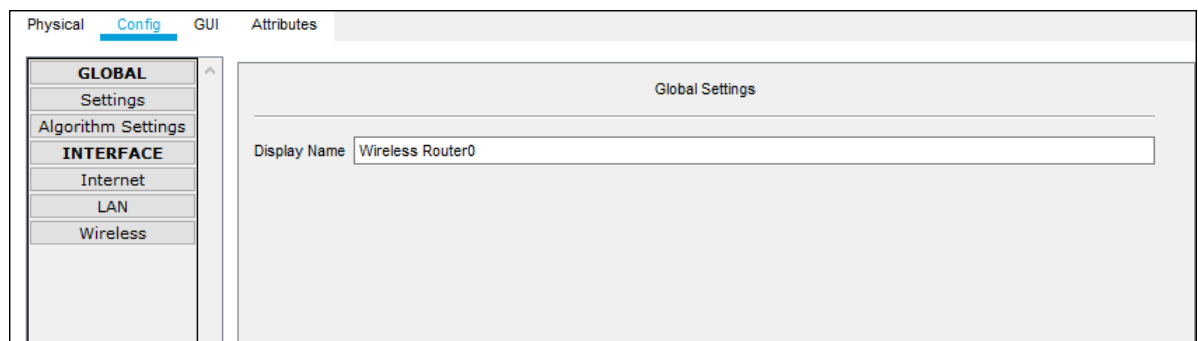
- b. Add network devices to the workspace.

Using the device selection box, add the network devices to the workspace as shown in the topology diagram

To place a device onto the workspace, first choose a device type from the **Device-Type Selection** box. Then, click on the desired device model from the **Device-Specific Selection** box. Finally, click on a location in the workspace to put your device in that location. If you want to cancel your selection, click the **Cancel** icon for that device. Alternatively, you can click and drag a device from the **Device-Specific Selection** box onto the workspace.

- c. Change display names of the network devices.

To change the display names of the network devices click on the device icon on the Packet Tracer **Logical** workspace, then click on the **Config** tab in the device configuration window. Type the new name of the device into the **Display Name** box as show in the figure below.



- d. Add the physical cabling between devices on the workspace

Using the device selection box, add the physical cabling between devices on the workspace as shown in the topology diagram.

The PC will need a copper straight-through cable to connect to the wireless router. Select the copper straight-through cable in the device selection box and attach it to the FastEthernet0 interface of the PC and the Ethernet 1 interface of the wireless router.

The wireless router will need a copper straight-through cable to connect to the cable modem. Select the copper straight-through cable in the device-selection box and attach it to the Internet interface of the wireless router and the Port 1 interface of the cable modem.

The cable modem will need a coaxial cable to connect to the Internet cloud. Select the coaxial cable in the device-selection box and attach it to the Port 0 interface of the cable modem and the coaxial interface of the Internet cloud.

The Internet cloud will need copper straight-through cable to connect to the Cisco.com server. Select the copper straight-through cable in the device-selection box and attach it to the Ethernet interface of the Internet cloud and the FastEthernet0 interface of the Cisco.com server.

Part 2: Configure the Network Devices

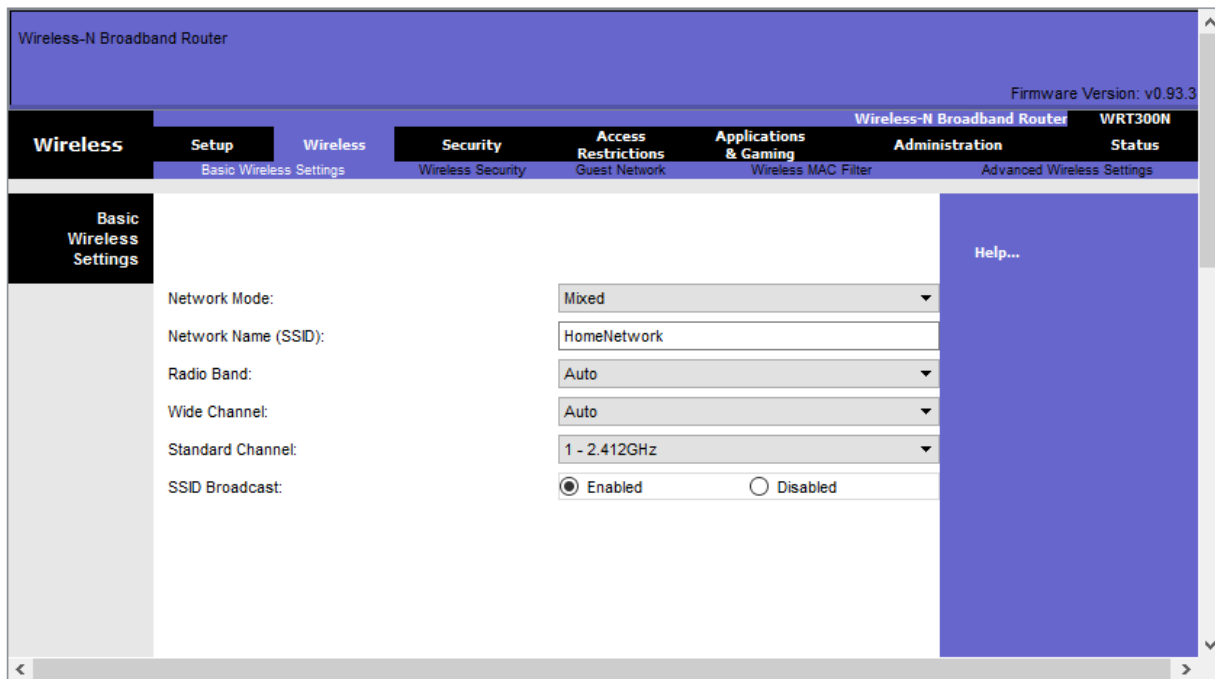
Step 1: Configure the wireless router

- a. Create the wireless network on the wireless router

Click on the **Wireless Router** icon on the Packet Tracer **Logical** workspace to open the device configuration window.

In the wireless router configuration window, click on the **GUI** tab to view configuration options for the wireless router.

Next, click on the **Wireless** tab in the GUI to view the wireless settings. The only setting that needs to be changed from the defaults is the **Network Name (SSID)**. Here, type the name “HomeNetwork” as shown in the figure.



Configure the Internet connection on the wireless router

Click on the **Setup** tab in the wireless router GUI.

In the **DHCP Server** settings verify that the **Enabled** button is selected and configure the static IP address of the DNS server as 208.67.220.220 as shown in the figure.

- b. Click on the **Save Settings** tab.

The screenshot shows the configuration interface for a Wireless-N Broadband Router (WRT300N). The top navigation bar includes tabs for Setup, Wireless, Security, Access Restrictions, Applications & Gaming, Administration, and Status. The 'Setup' tab is active, and the 'Network Setup' sub-tab is selected. On the left, a sidebar lists 'Internet Setup' and 'Network Setup'. The 'Network Setup' section is expanded, showing 'Router IP' and 'DHCP Server Settings'. The 'DHCP Server Settings' are configured as follows:

- Internet Connection type:** Automatic Configuration - DHCP
- Host Name:** (empty field)
- Domain Name:** (empty field)
- MTU:** 1500
- IP Address:** 192.168.0.1
- Subnet Mask:** 255.255.255.0
- DHCP Server:** Enabled (radio button selected)
- Start IP Address:** 192.168.0.100
- Maximum number of Users:** 50
- IP Address Range:** 192.168.0.100 - 149
- Client Lease Time:** 0 minutes (0 means one day)
- Static DNS 1:** 208.67.220.220
- Static DNS 2:** 0.0.0.0
- Static DNS 3:** 0.0.0.0
- WINS:** 0.0.0.0

Step 2: Configure the laptop

- a. Configure the Laptop to access the wireless network

Click on the Laptop icon on the Packet Tracer **Logical** workspace and in the laptop configuration windows select the **Physical** tab.

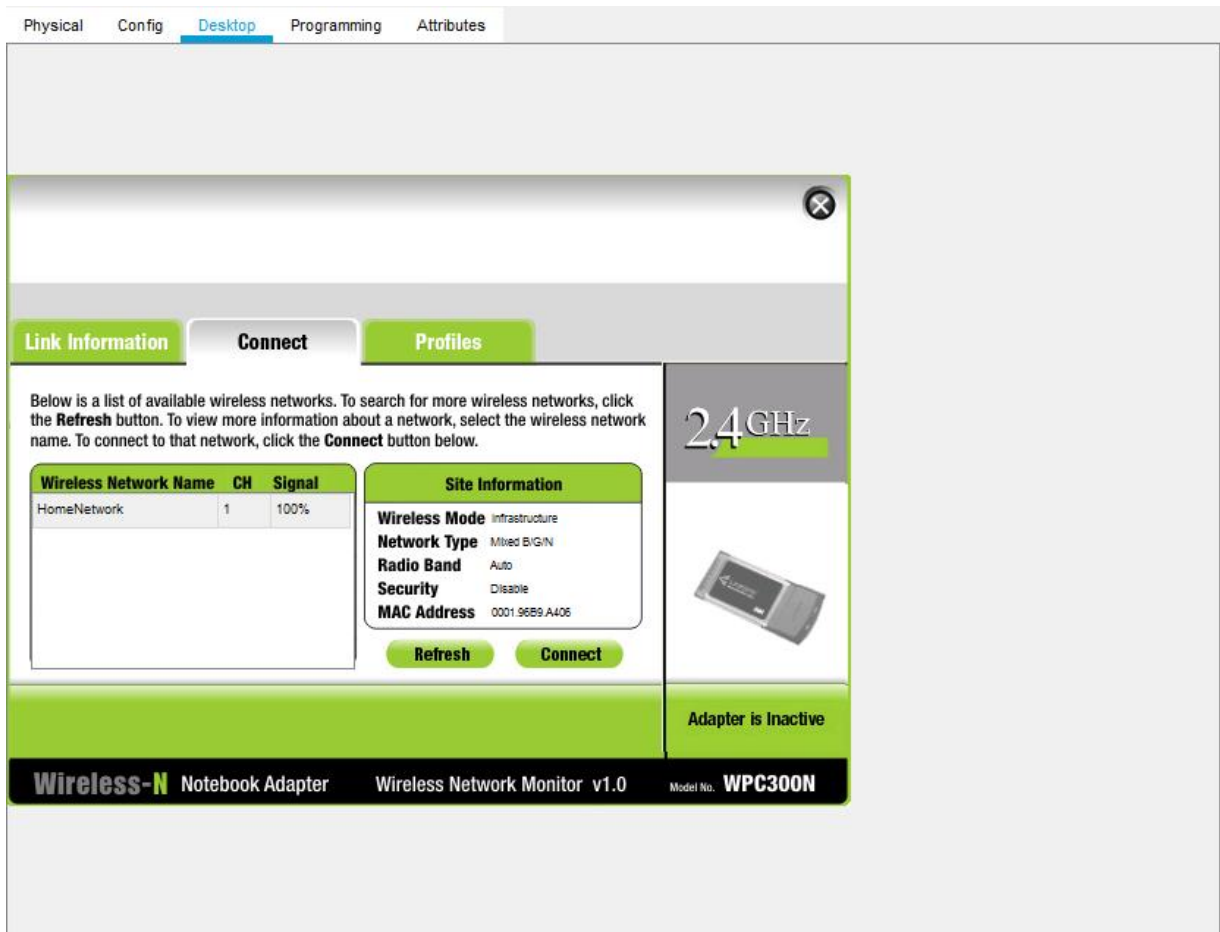
In the **Physical** tab you will need to remove the Ethernet copper module and replace it with the Wireless WPC300N module.

To do this, you first power the Laptop off by clicking the power button on the side of the laptop. Then remove the currently installed Ethernet copper module by clicking on the module on the side of the laptop and dragging it to the **MODULES** pane on the left of the laptop window. Then install the Wireless WPC300N module by clicking on it in the **MODULES** pane and dragging it to the empty module port on the side of the laptop. Power the laptop back on by clicking on the Laptop power button again.

With the wireless module installed, the next task is to connect the laptop to the wireless network.

Click on the **Desktop** tab at the top of the Laptop configuration window and select the **PC Wireless** icon. Once the Wireless-N Notebook Adapter settings are visible, select the **Connect** tab. The wireless network “HomeNetwork” should be visible in the list of wireless networks as shown in the figure.

Select the network, and click on the **Connect** tab found below the **Site Information** pane.

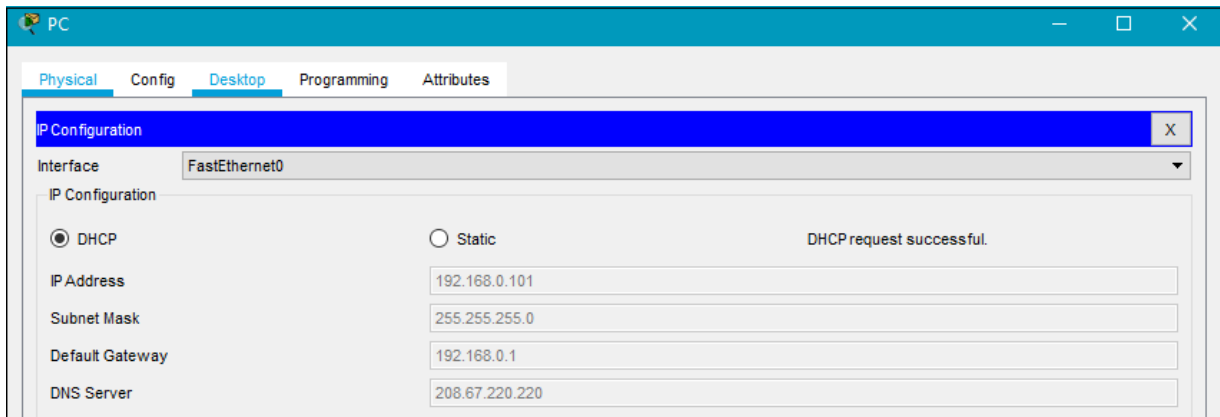


Step 3: Configure the PC

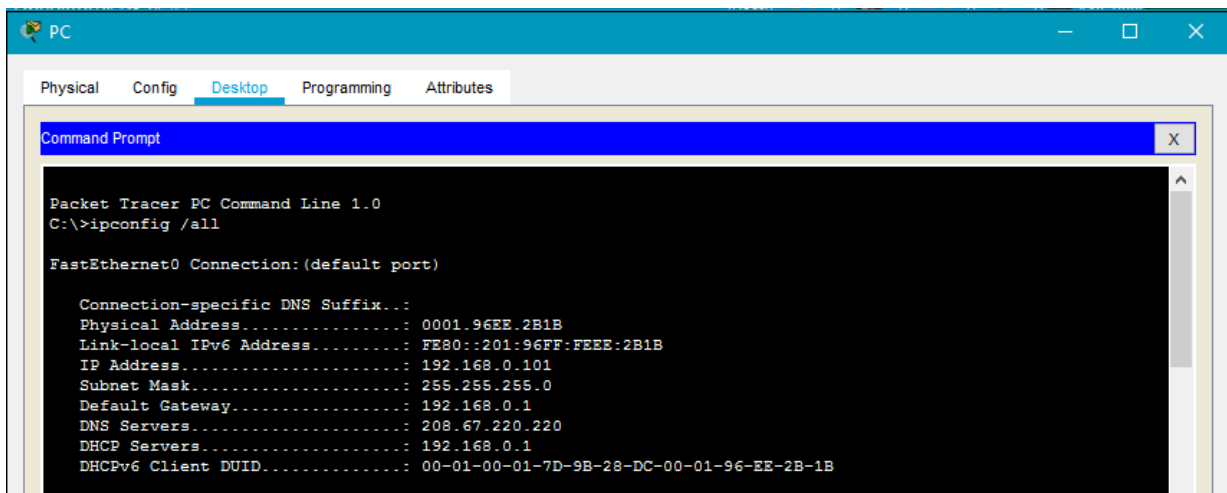
- Configure the PC for the wired network

Click on the **PC** icon on the Packet Tracer **Logical** workspace and select the **Desktop** tab and then the **IP Configuration** icon.

In the IP Configuration window, select the **DCHP** radio button as shown in the figure so that the PC will use DHCP to receive an IPv4 address from the wireless router. Close the IP Configuration window.



Click on the Command Prompt icon. Verify that the PC has received an IPv4 address by issuing the **ipconfig /all** command from the command prompt as shown in the figure. The PC should receive an IPv4 address in the 192.168.0.x range.



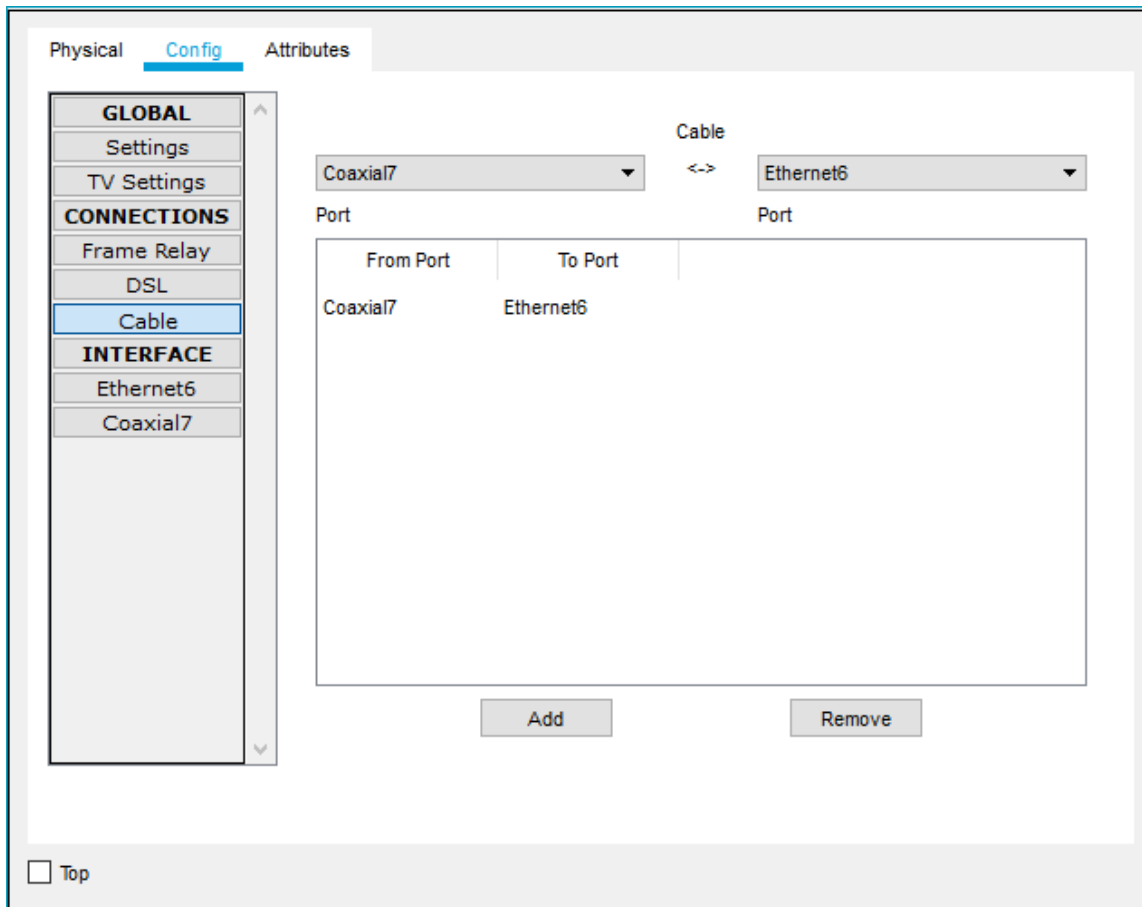
Step 4: Configure the Internet cloud

- a. Install network modules if necessary

Click on the **Internet Cloud** icon on the Packet Tracer **Logical** workspace and then click on the **Physical** tab. The cloud device will need two modules if they are not already installed. The PT-CLOUD-NM-1CX which is for the cable modem service connection and the PT-CLOUD-NM-1CFE which is for a copper Ethernet cable connection. If these modules are missing, power off the physical cloud devices by clicking on the power button and drag each module to an empty module port on the device and then power the device back on.

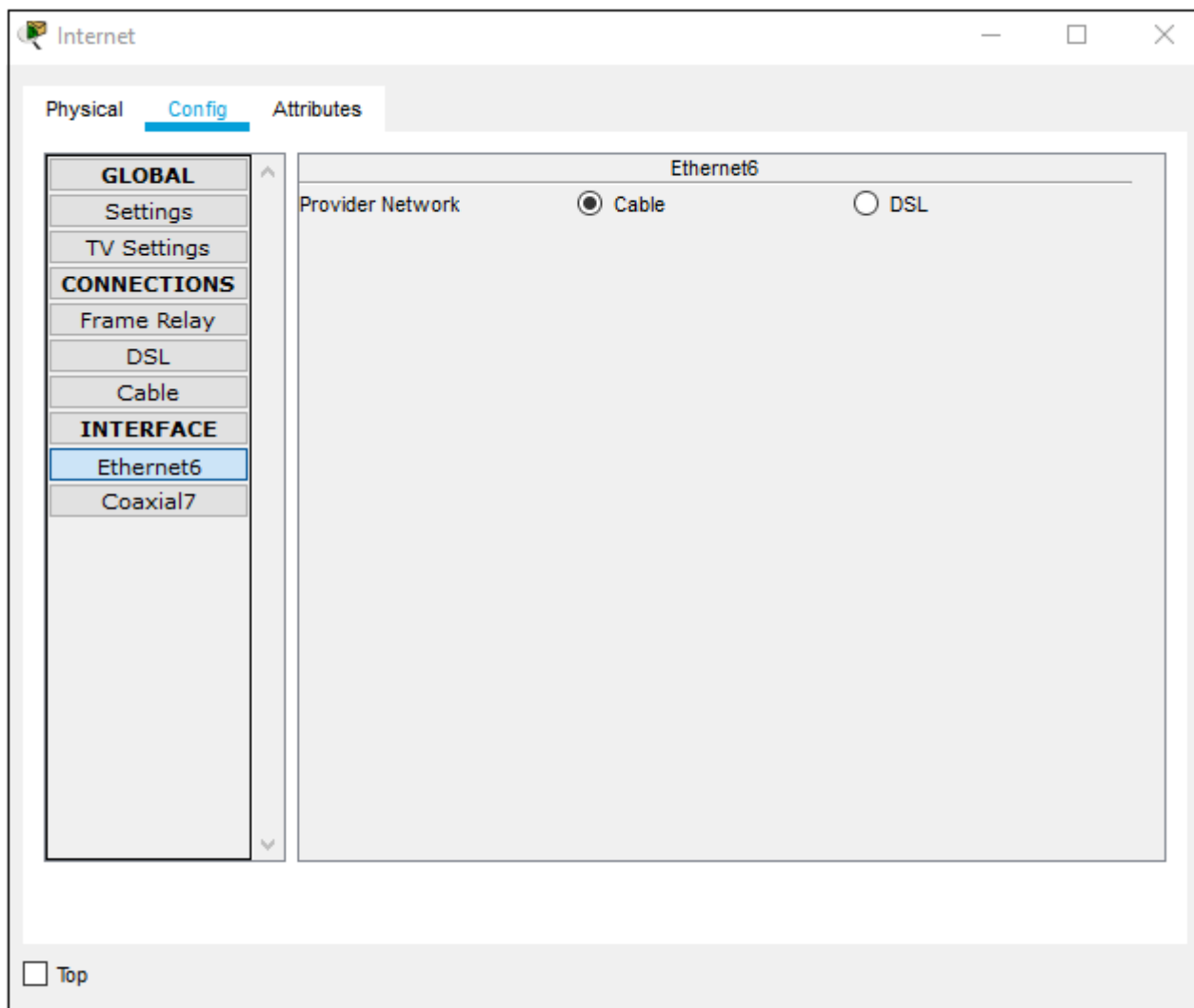
- b. Identify the From and To Ports

Click on the **Config** tab in the Cloud device window. In the left pane click on **Cable** under **CONNECTIONS**. In the first drop down box choose Coaxial and in the second drop down box choose Ethernet then click the **Add** button to add these as the **From Port** and **To Port** as shown in the figure.



c. Identify the type of provider

While still in the **Config** tab click Ethernet under **INTERFACE** in the left pane. In the Ethernet configuration window select **Cable** as the Provider Network as shown in the figure.



Step 5: Configure the Cisco.com server

- a. Configure the Cisco.com server as a DHCP server

Click on the Cisco.com server icon on the Packet Tracer **Logical** workspace and select the **Services** tab. Select **DHCP** from the **SERVICES** list in the left pane.

In the DHCP configuration window, configure a DHCP as shown in the figure with the following settings.

- Click **On** to turn the DHCP service on
- Pool name: DHCPpool
- Default Gateway: 208.67.220.220
- DNS Server: 208.67.220.220
- Starting IP Address: 208.67.220.1
- Subnet Mask 255.255.255.0
- Maximum number of Users: 50

Click **Add** to add the pool

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: DHCPpool

Default Gateway: 208.67.220.220

DNS Server: 208.67.220.220

Start IP Address : 208 67 220 1

Subnet Mask: 255 255 255 0

Maximum Number of Users : 50

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Buttons: Add Save Remove

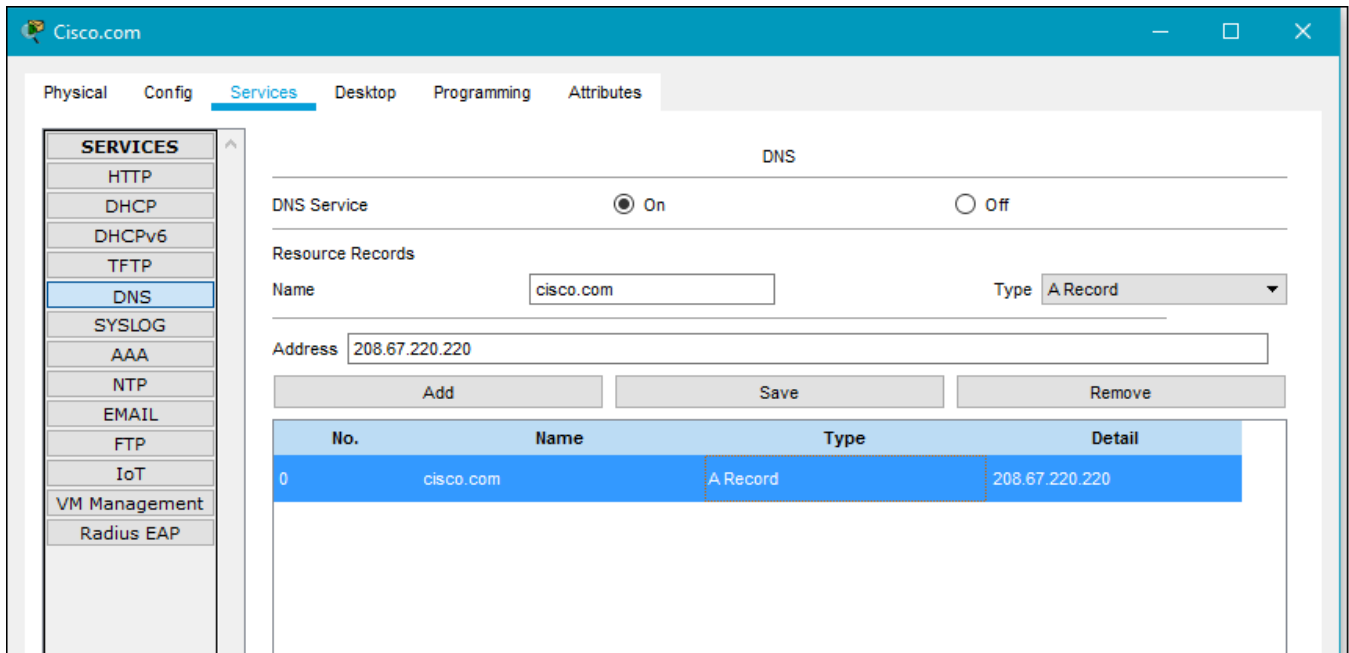
Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
DHCPpool	208.67.220.220	208.67.220.220	208.67.220.1	255.255.255.0	50	0.0.0.0	0.0.0.0

- b. Configure the Cisco.com server as a DNS server to provide domain name to IPv4 address resolution. While still in the **Services** tab, select **DNS** from the **SERVICES** listed in the left pane.

Configure the DNS service using the following settings as shown in the figure.

- Click **On** to turn the DNS service on
- Name: Cisco.com
- Type: A Record
- Address: 208.67.220.220

Click **Add** to add the DNS service settings



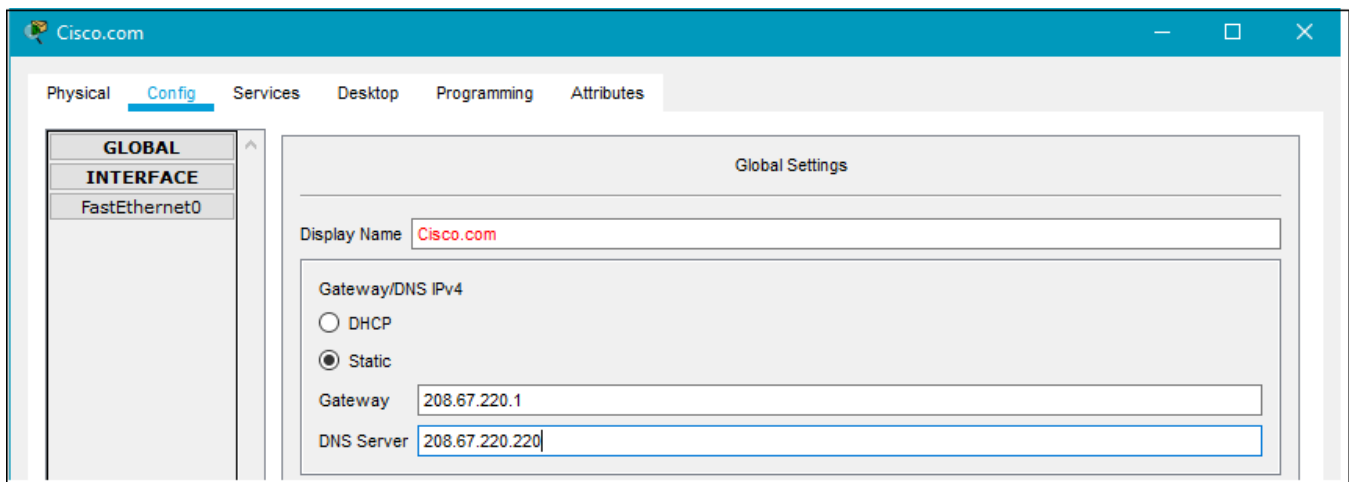
c. Configure the Cisco.com server Global settings.

Select the **Config** tab.

Click on **Settings** in left pane.

Configure the Global settings of the server as follows:

- Select **Static**
- Gateway: 208.67.220.1
- DNS Server: 208.67.220.220



d. Configure the Cisco.com server FastEthernet0 Interface settings.

Click on **FastEthernet** in left pane of the **Config** tab

Configure the FastEthernet Interface settings of the server as follows:

- Select **Static** under IP Configuration
- IP Address: 208.67.220.220
- Subnet Mask: 255.255.255.0

The screenshot shows the Cisco Packet Tracer configuration window for the FastEthernet0 interface. The window has a blue header with the Cisco logo and the text "Cisco.com". Below the header is a navigation bar with tabs: Physical, Config (selected), Services, Desktop, Programming, and Attributes. On the left side, there is a sidebar with a tree view showing "GLOBAL" and "INTERFACE" categories. Under "INTERFACE", "FastEthernet0" is selected. The main area displays the configuration for FastEthernet0. It includes fields for Port Status (checked On), Bandwidth (10 Mbps selected), Duplex (Full Duplex selected), and MAC Address (000C.85CC.0116). Below these are sections for IP Configuration and IPv6 Configuration. In the IP Configuration section, "Static" is selected, and the IP Address is set to 208.67.220.220 and the Subnet Mask is set to 255.255.255.0. In the IPv6 Configuration section, "Static" is selected, and the IPv6 Address is empty. The Link Local Address is set to FE80::20C:85FF:FECC:116. At the bottom left, there is a "Top" button.

Part 3: Verify Connectivity

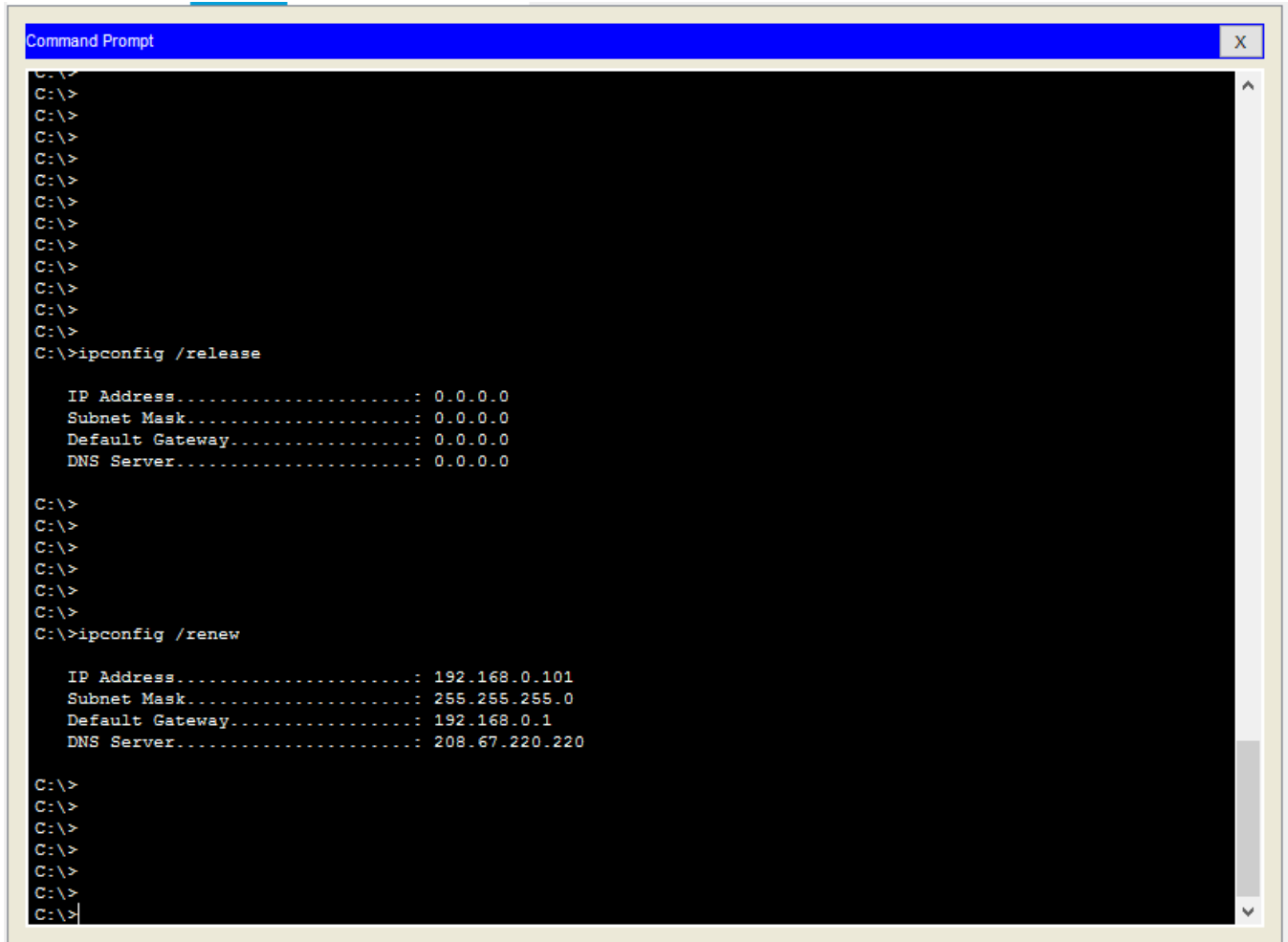
Step 1: Refresh the IPv4 settings on the PC

- Verify that the PC is receiving IPv4 configuration information from DHCP.

Click on the **PC** on the Packet Tracer **Logical** workspace and then select the **Desktop** tab of the PC configuration window.

Click on the **Command Prompt** icon

In the command prompt refresh the IP settings by issuing the commands **ipconfig /release** and then **ipconfig /renew**. The output should show that the PC has an IP address in the 192.168.0.x range, a subnet mask, a default gateway, and DNS server address as shown in the figure.

A screenshot of a Windows Command Prompt window. The title bar is blue and says "Command Prompt". The background is black with white text. The prompt is "C:\>". The user has entered "ipconfig /release" and the output shows all IP-related fields set to 0.0.0.0. Then the user enters "ipconfig /renew" and the output shows a new IP address of 192.168.0.101, subnet mask 255.255.255.0, default gateway 192.168.0.1, and DNS server 208.67.220.220.

```
Command Prompt
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>ipconfig /release

IP Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway...: 0.0.0.0
DNS Server.....: 0.0.0.0

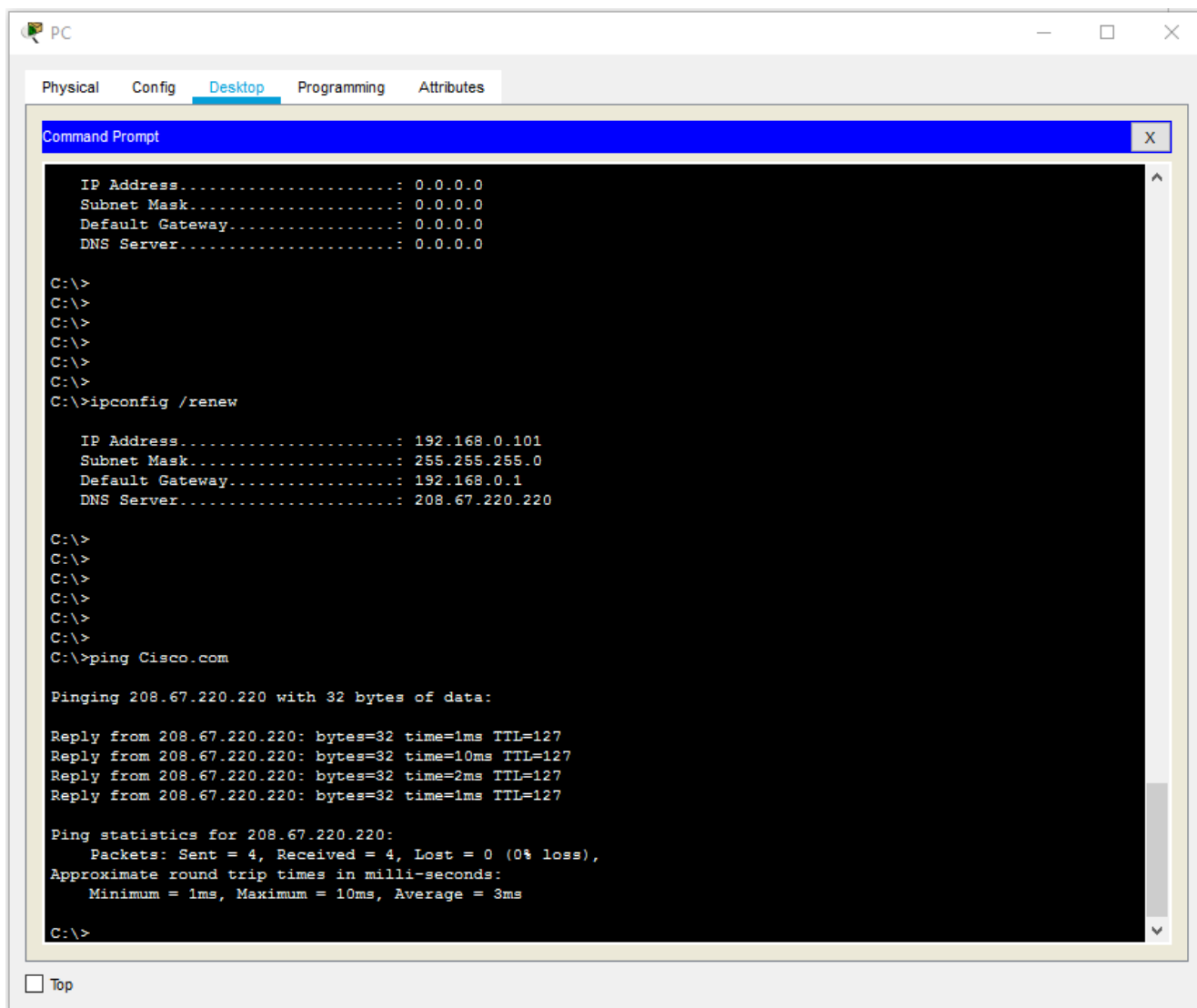
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>ipconfig /renew

IP Address.....: 192.168.0.101
Subnet Mask.....: 255.255.255.0
Default Gateway...: 192.168.0.1
DNS Server.....: 208.67.220.220

C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
```

- b) Test connectivity to the Cisco.com server from the PC

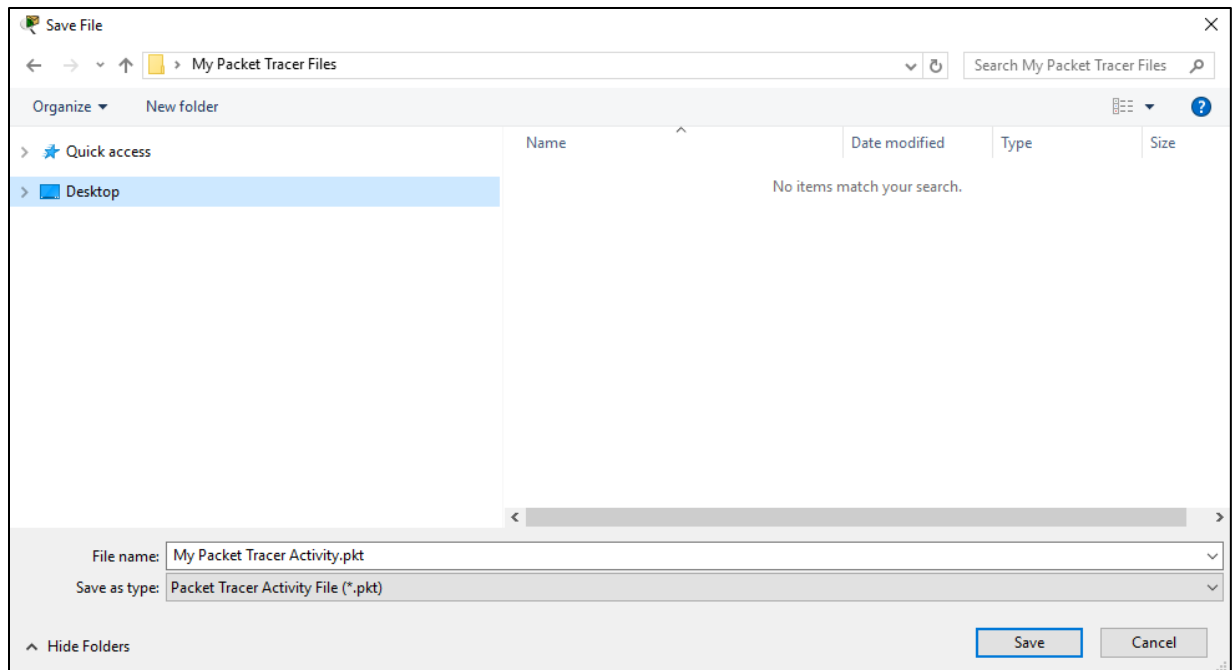
From the command prompt, issue the command **ping Cisco.com**. It may take a few seconds for the ping to return. Four replies should be received as shown in the figure.



Part 4: Save the File and Close Packet Tracer

Step 1: Save the File as a Packet Tracer Activity File (*.pkt)

To save the completed network, click on **File** in the Packet Tracer menu bar and then select **Save A** from the dropdown menu. In the **Save File** window choose a directory to save the file to and give the file an appropriate file name. The Save as type defaults to Packet Tracer Activity File (*.pkt). Click **Save** to save the file.



Step 2: Close Packet Tracer

To close Packet Tracer you can either click the “X” in the top right corner of the Packet Tracer window, or click on **Exit** in the file drop down menu.

Lab Tasks

In-Lab Task 1:

- Implement all steps mentioned in lab manual.
- Write all steps implemented in lab in summarized form.

Post lab task 1:

Attach screenshots of all steps implemented in lab.